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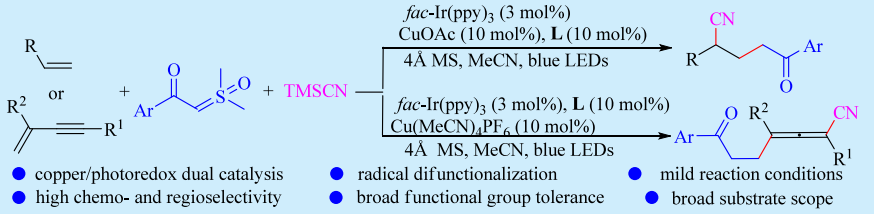
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Dual Photoredox/Copper-Catalyzed Three-Component Alkylcyanation of Alkenes and 1,4-Alkylcyanation of 1,3-Enynes Employing Sulfoxonium Ylides as the Carbon Radical Precursors

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ABSTRACT: A novel dual photoredox/copper-catalyzed three-component alkylcyanation of alkenes and 1,4-alkylcyanation of 1,3-enynes have been developed. In this radical cyanoalkylation reaction, the photoredox induced alkyl radical from sulfoxonium ylides adds to the carbon−carbon double bonds of styrenes or 1,3-enynes, and the generated benzylic or allenyl radicals couple with a Cu(II) cyanide complex to achieve selective cyanation. The reaction exhibits high chemo- and regioselectivity and a wide substrate scope, providing an efficient method for the synthesis of alkyl nitriles and allenyl nitriles in a single step.

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Alkenes have proven to be ideal building blocks in organic chemistry due to their easy accessibility, abundance and versatile reactivity.[1] Among various transformations of alkenes, the difunctionalization of alkenes has been widely developed, which allows the introduction of two functional groups to the double bond of alkenes in a single step.[2] In particular, the intermolecular three-component radical difunctionalization of alkenes has become a powerful strategy for efficiently enhancing molecular complexity.[3] Recently, the visible-light-induced radical strategy has further matured the radical difunctionalization of alkenes under environmentally benign mild conditions.[4] In this field, the dual photoredox/copper-catalyzed cyanoalkylation of alkenes has been reported.[5,6] However, it still faces challenges due to the limited availability of alkyl radical precursors and only *N*-hydroxyl phthalimide esters[5] and alkyl bromides[6] can be used as carbon radical sources. In addition, sulfur ylides are regarded as a kind of versatile building block in organic synthesis due to their rich reactivity, high stability and easy preparation.[7] With the development of photochemistry, visible-light-induced photoredox carbon radical formation from sulfur ylides via homolytic cleavage of C−S bonds has received considerable attention in recent years.[8] However, to the best of our knowledge, the visible-light-induced cyanoalkylation of alkenes with carbon radicals generated from sulfoxonium ylides has not been reported. Thus, realizing the dual photoredox/metal-catalyzed cyanoalkylation of alkenes with novel alkyl radical precursors, especially for the radical cyanoalkylation of alkenes with sulfoxonium ylides and TMSCN, is highly desired.

Allenes represent important structural units found frequently in many natural products, pharmaceuticals, and molecular materials.[9] Moreover, they are also useful and versatile building blocks for the synthesis of diverse and valuable compounds in organic synthesis.[10] Consequently, many methods for the synthesis of allenes have been developed.[11] Among these, the 1,4-difunctionalization of 1,3-enynes is considered to be one of the best methods to construct multisubstituted allenes because two functionalities are installed in a single step.[12−19] In recent years, there has been remarkable progress in photoredox- and transition metal-catalyzed radical 1,4-difunctionalization of 1,3-enynes for the synthesis of allenes.[13−19] In this context, the radical-mediated 1,4-alkylcyanation of 1,3-enynes employing diacyl peroxides,[14] fluoroalkyl halides,[15] *N*-hydroxyphthalimide (NHP) esters,[16] Togni reagents,[17] cyclo-butanone oxime esters,[18] [difluoro(phenylthio)methyl]triphenylphosphonium triflate/(difluoromethyl)triphenylphosphonium bromide,[19] and cyclic alcohol derivatives[20] as carbon radical sources has been reported for the synthesis of multiple-functionalized allenyl nitriles. Therefore, the development of visible-light-induced 1,4-alkylcyanation of 1,3-enynes with novel carbon radical precursors for synthesizing diverse allenyl nitriles is in continuous demand. As our continued interest in the applications of alkynes[21] and visible-light-promoted radical

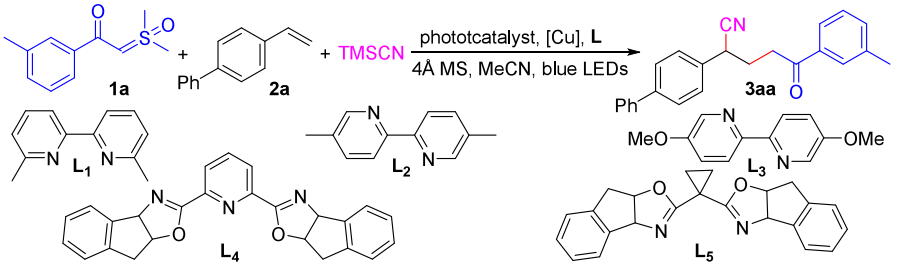
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Table 1. Optimization of Reaction Conditions[a,b]



entry	cat. (mol %)	ligand (mol)	[Cu] (mol %)	solvent	yield (%)
1	*fac*-Ir(ppy)$_3$ (1)	L_4 (10)	CuOAc (10)	MeCN	70
2	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	CuOAc (10)	MeCN	81/76[c]/68[d]/34[e]
3	[Ir(dtbbpy)(ppy)$_2$]PF$_6$ (3)	L_4 (10)	CuOAc (10)	MeCN	49
4	Ru(bpy)$_3$Cl$_2$·6H$_2$O (3)	L_4 (10)	CuOAc (10)	MeCN	19
5	Ru(bpy)$_3$(PF$_6$)$_2$ (3)	L_4 (10)	CuOAc (10)	MeCN	24
6	4CzIPN (3)	L_4 (10)	CuOAc (10)	MeCN	trace
7	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	CuOAc (3)	MeCN	70
8	*fac*-Ir(ppy)$_3$ (3)	L_4 (5)	CuOAc (15)	MeCN	73
9	*fac*-Ir(ppy)$_3$ (3)	L_1 (10)	CuOAc (10)	MeCN	62
10	*fac*-Ir(ppy)$_3$ (3)	L_2 (10)	CuOAc (10)	MeCN	40
11	*fac*-Ir(ppy)$_3$ (3)	L_3 (10)	CuOAc (10)	MeCN	44
12	*fac*-Ir(ppy)$_3$ (3)	L_5 (10)	CuOAc (10)	MeCN	39
13	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	Cu(OAc)$_2$ (10)	MeCN	74
14	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	Cu(OTf)$_2$ (10)	MeCN	71
15	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	CuBr (10)	MeCN	56
16	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	Cu(CH$_3$CN)$_4$PF$_6$ (10)	MeCN	trace
17	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	CuOAc (10)	DCM	42
18	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	CuOAc (10)	DMF	31
19	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	CuOAc (10)	THF	39
20	*fac*-Ir(ppy)$_3$ (3)	L_4 (10)	CuOAc (10)	PhCl	trace

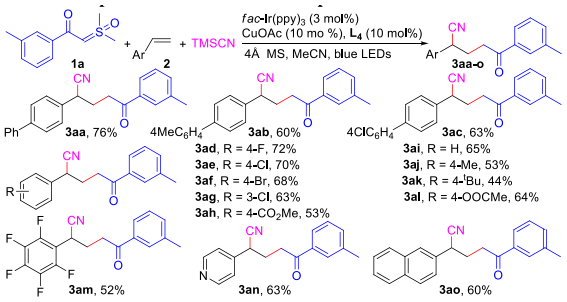
[a]Reaction conditions: **1a** (0.6 mmol), **2a** (0.2 mmol), TMSCN (0.6 mmol), *fac*-Ir(ppy)$_3$ (1−3 mol %), CuOAc (3−15 mol %), L_4 (5−10 mol %), 4 Å MS (60.0 mg), MeCN (2.0 mL), blue LEDs (20 W), under N_2, at room temperature for 24 h. [b]Yields were determined by ^{1}H NMR spectroscooy using 1,3,5-trimethoxybenzene as the internal standard. [c]Isolated yield. [d]Yield with 0.4 mmol of **1a**. [e]Yield with 0.4 mmol of TMSCN.

reactions,[22] herein we report a novel dual photoredox/copper-catalyzed three-component alkyl-cyanation of alkenes and 1,4-alkylcyanation of 1,3-enynes employing sulfoxonium ylides as carbon radical sources. The reaction exhibits high chemo- and regioselectivity and a wide substrate scope and provides an efficient route to the synthesis of carbonyl-containing alkyl nitriles and allenyl nitriles in a single step.

In our initial study, the reaction of sulfoxonium ylide **1a**, 4-vinylbiphenyl **2a** and trimethylsilanecarbonitrile TMSCN was selected as a model reaction to optimize the reaction conditions (Table 1). We found that the visible-light-induced cyanoalkylation of **2a** (0.2 mmol), **1a** (0.6 mmol) and TMSCN (0.6 mmol) proceeded smoothly to give alkyl nitrile **3aa** in 70% yield in the presence of *fac*-Ir(ppy)$_3$ (1 mol %), CuOAc (10 mol %), ligand L_4 (10 mol %) and 4 Å molecular sieves (60 mg) under irradiation of a 20 W LEDs in acetonitrile (2 mL) at room temperature for 24 h (entry 1). Gratifyingly, the yield of **3aa** was raised to 76% in the presence of *fac*-Ir(ppy)$_3$ (3 mol %) (entry 2). Decreasing the amount of CuOAc or L_4 led to comparatively lower yields of **3aa** (entries 7 and 8). When Cu(OAc)$_2$ and Cu(OTf)$_2$ were employed as catalysts, product **3aa** was obtained in 74% and 71% yields, respectively (entries 13 and 14). Other photocatalysts, ligands, and copper salts, such as [Ir(dtbbpy)(ppy)$_2$]PF$_6$, Ru(bpy)$_3$Cl$_2$·6H$_2$O, Ru(bpy)$_3$(PF$_6$)$_2$, 4CzIPN, L_1-$_3$, L_5, CuBr and Cu(MeCN)$_4$PF$_6$ were less effective (entries 3−5, 9−12 and 15) or ineffective (entries 6 and 16). Further evaluation of solvents revealed that CH$_3$CN was superior to other solvents, such as DCM, DMF, and THF (entries 17−19) and the reaction occurred difficultly in PhCl (entry 20).

With the optimized reaction conditions in hand, the scope of the carbocyanation reaction with respect to alkenes was evaluated and the results are summarized in Table 2. Various selected styrene derivatives **2a**−**l**, bearing either electron-

Table 2. Scope of the Reaction with Respect to the Alkenes 2[a,b]



[a]Reaction conditions: **1a** (0.6 mmol), **2** (0.2 mmol), TMSCN (0.6 mmol), *fac*-Ir(ppy)$_3$ (3 mol %), CuOAc (10 mol %), L_4 (10 mol %), 4 Å MS (60.0 mg), MeCN (2.0 mL), blue LEDs (20 W), under N_2, at room temperature for 24 h. [b]Isolated yield.

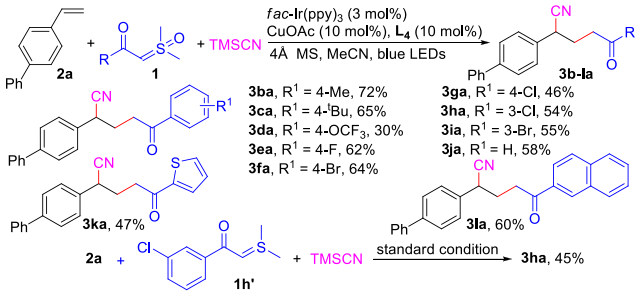
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withdrawing (Ar, Cl, Br, F, and CO_2Me) or electron-donating groups (methyl, *tert*-butyl, and OCOMe) on the phenyl ring, could react smoothly with **1a** and TMSCN to produce the corresponding carbonyl-containing alkyl nitriles **3aa−l** in good to high yields. Similarly, pentafluorostyrene **2m** was also compatible with this cyanoalkylation reaction and gave the desired product **3am** in 52% yield. Furthermore, the difunctionalization proceeded well with 4-vinylpyridine **2n** and 2-vinylnaphthalene **2o**, with the generation of alkyl nitriles **3an** and **3ao** in 63% and 60% yields, respectively. However, no desired products were observed when pent-4-en-1-ylbenzene, 2-bromobut-1-ene or ethene-1,1-diyldibenzene was used as a coupling partner. A scale-up reaction of **1a** (21.0 mmol), **2a** (7.0 mmol) and TMSCN (21.0 mmol) was carried out for 48 h under otherwise identical conditions, affording 1.76 g of the product **3aa** in 74% yield.

Next, the substrate scope of sulfoxonium ylides **1** was evaluated. As shown in Table 3, various benzoyl-substituted sulfoxonium ylides **1b−j** bearing electron-donating, electron-withdrawing and electron-neutral substituents on the aromatic ring, could react with **2a** and TMSCN to give the corresponding alkyl nitriles **3b−ja** in 30−72% yields. In this transformation, a variety of functional groups, such as alkyl, trifluoromethoxy, fluorine, chlorine, and bromine, were tolerated. Moreover, this reaction proceeded well with heteroaryl-substituted sulfoxonium ylide **1k** and 2-naphthyl-substituted sulfoxonium ylide **1l** to deliver desired products **3ka** and **3la** in 47% and 60% yields, respectively. Notably, even employing sulfur ylide **1h′** as a carbon radical source, the desired product **3ha** was obtained in 45% yield. However, no desired product was obtained when cyclohexylacyl-substituted sulfoxonium ylide was employed as the radical precursor.

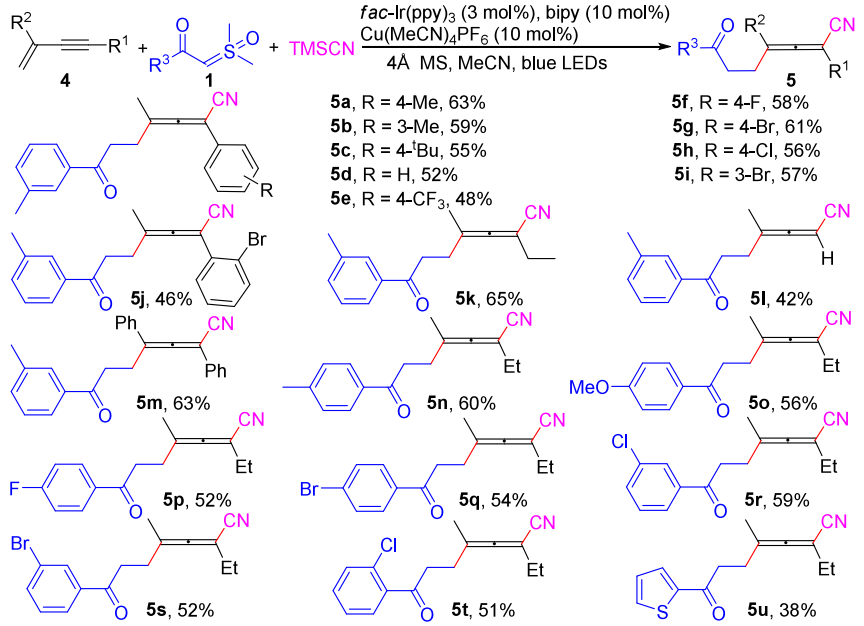
To further explore the generality of this difunctionalization of alkenes, the visible light-induced 1,4-alkylcyanation of 1,3-enynes **4** with sulfoxonium ylides **1** and TMSCN was investigated and the results are summarized in Table 4. After a series of optimization steps, we found that the radical-mediated three-component 1,4-alkylcyanation of 4-aryl-1,3-enyne **4a** (0.2 mmol) with sulfoxonium ylide **1a** (0.6 mmol) and TMSCN (0.6 mmol) proceeded also to give the allenyl nitrile **5a** in 63% yield in the presence of *fac*-Ir(ppy)$_3$ (3 mol %), Cu(MeCN)$_4$PF$_6$ (10 mol %), and 2,2′-bipy (10 mol %) under other identical conditions. Similarly, 4-aryl-1,3-enynes **4b−j** bearing electron-donating (Me, ^{t}Bu), electron-withdrawing (CF_3, F, Cl, Br) and electron-neutral substituents on the phenyl ring were found to be compatible with the 1,4-alkylcyanation reaction, provided the corresponding allenyl nitriles **5b−j** in moderate to good yields. Moreover, the 1,4-alkylcyanation reaction proceeded well with 4-alkyl-1,3-enyne **4k** to deliver product **5k** in 65% yield. Notably, 2-methylbut-1-en-3-yne **4l** also proved to be the suitable partner for this reaction, affording the desired product **5l** in 42% yield. More importantly, even using but-3-en-1-yne-1,3-diyldibenzene **4m** as the coupling partner, the reaction also worked well, with the generation of allenyl nitrile **5m** in 63% yield. In addition,

Table 3. Scope of the Sulfoxonium Ylides 1[a,b]



[a]Reaction conditions: **1** (0.6 mmol), **2a** (0.2 mmol), TMSCN (0.6 mmol), *fac*-Ir(ppy)$_3$ (3 mol %), CuOAc (10 mol %), **L$_4$** (10 mol %), 4 Å MS (60.0 mg), MeCN (2.0 mL), blue LEDs (20 W), under N_2, at room temperature for 24 h. [b]Isolated yield.

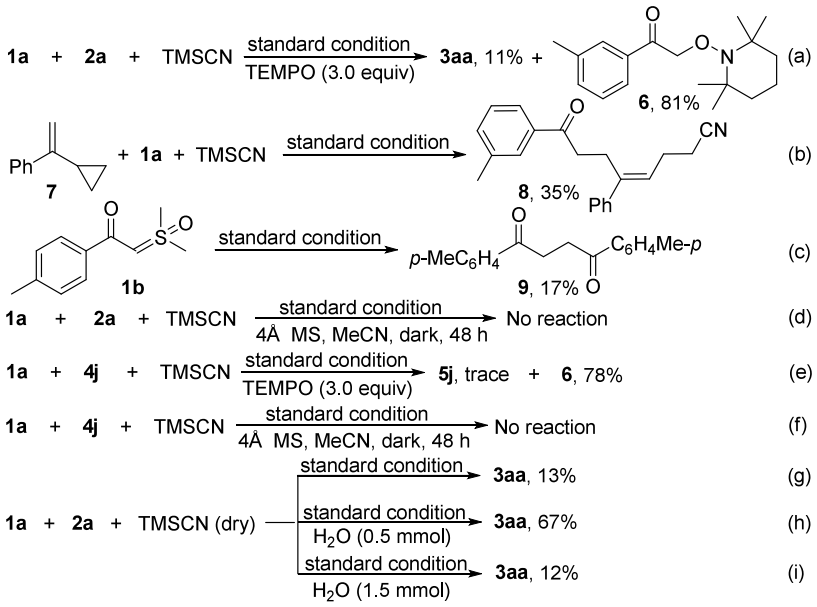
Table 4. 1,4-Alkylcyanation of 1,3-Enynes 4[a,b]



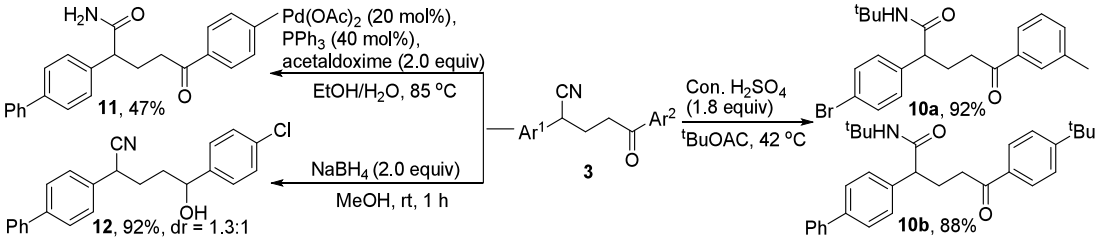
[a]Reaction conditions: **1** (0.6 mmol), **4** (0.2 mmol), TMSCN (0.6 mmol), *fac*-Ir(ppy)$_3$ (3 mol %), Cu(MeCN)$_4$PF$_6$ (10 mol %), 2,2′-bipy (10 mol %), 4 Å MS (60.0 mg), MeCN (2.0 mL), blue LEDs (20 W), under N_2, at room temperature for 24 h. [b]Isolated yield.

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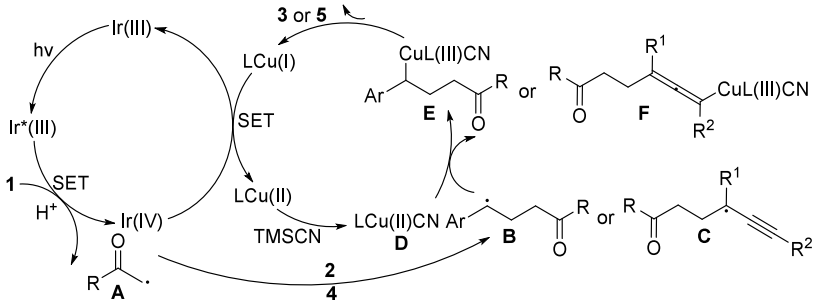
Scheme 1. Mechanistic Studies



Scheme 2. Potential Synthetic Applications of 3



Scheme 3. Proposed Mechanisms for the Formation of 3 and 5



various benzoyl-substituted sulfoxonium ylides **1** bearing electron-donating and electron-withdrawing substituents on the phenyl ring, could react smoothly with **4k** and TMSCN to give the corresponding allenyl nitriles **5n−t** in good yields. Similarly, heteroaryl-substituted sulfoxonium ylide **1k** was also tolerated and the product **5u** was produced in 38% yield.

To further probe the reaction mechanisms, some control experiments were investigated (Scheme 1). We found that the formation of **3aa** was drastically reduced from 76% to 11% in the presence of 2,2,6,6-tetramethylpiperidinooxy (TEMPO, 3.0 equiv) under the optimized condition, whereas the TEMPO-trapped-carbonyl carbon radical compound **6** was obtained in 81% yield ((a)). When the reaction of **1a**, TMSCN and (1-cyclopropylvinyl)benzene **7** as a radical clock was performed under the standard conditions, the product **8** was obtained in 35% yield ((b)). The product **9** generated by the self-coupling of carbonyl carbon radical **A** (Scheme 3) was formed in 17% yield when **1b** was treated under the standard condition ((c)). As anticipated, visible light irradiation is critical for this alkylcyanation of alkene **2a** and 1,4-alkylcyanation of 1,3-enyne **4j** ((d and f)). Similarly, only a trace amount of **5j** was observed for the 1,4-alkylcyanation reaction of **1a**, **4j** and TMSCN in the presence of TEMPO (3.0 equiv) and product **6** was also produced in 78% yield ((e)). These experimental results demonstrate that the alkylcyanation reaction might be a radical process and that carbonyl carbon radical **A** might be generated during the reaction. The product **3aa** was obtained in 13% yield when dry TMSCN was used as substrate ((g)). When 2.5 or 7.5 equiv of H_2O was added to the reaction system, the product **3aa** was generated in 67% and 12% yields ((h) and (i)), respectively. These experimental results imply that water (as proton source) is crucial for this reaction, but excessive water is

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detrimental to it. According to the linear Stern−Volmer behavior, sulfoxonium ylide **1a** more easily quench the excited photosensitizer than **2a** do (see the Supporting Information).

On the basis of the experimental results and related literature,[4−6,8,14−20] the plausible reaction mechanism for the dual photoredox/copper-catalyzed cyanoalkylation of alkenes or 1,3-enynes is proposed in Scheme 3. Initially, the photocatalyst *fac*-Ir(ppy)$_3$ absorbs visible light and produces the excited state (*fac*-Ir(ppy)$_3$)* under light irradiation, which is oxidatively quenched by **1** via an SET process to give carbonyl carbon radical **A** and *fac*-Ir(ppy)$_3$(IV). Subsequently, addition of intermediate **A** to alkenes **2** or 1,3-enynes **4** leads to the formation of carbon radical **B** or **C**. Meanwhile, the Cu(I) species is oxidized by high-valence *fac*-Ir(ppy)$_3$(IV) to afford Cu(II) species, which could react with TMSCN by ligand exchange to furnish the Cu(II) complex **D**. Then, carbon radical **B** or **C** is captured by intermediate **D** to deliver high-valent Cu(III) complex **E** or **F**. Finally, intermediate **E** or **F** undergoes reductive elimination to produce products **3** or **5** with regeneration of the LCu(I) complex and completes the copper catalysis cycle.

Furthermore, the synthetic utility of alkyl nitriles **3** was investigated. We found that the alkyl nitriles **3af** and **3ca** could be easily transformed to *N*-*tert*-butyl amides **10a** and **10b** in 92% and 88% yields in the presence of H_2SO_4 in ^{t}BuOAc, respectively. Transformation of **3ba** to the amide **11** was performed by treatment with (*E*/*Z*)-acetaldoxime under Pd catalysis in 47% yield. In addition, **3ga** was easily reduced to give secondary alcohol **12** in 92% yield in the presence of $NaBH_4$ in MeOH with 1.3:1 dr (Scheme 2).

In conclusion, we have developed a novel dual photoredox- and copper-catalyzed three-component alkylcyanation of alkenes with TMSCN and sulfoxonium ylides for the first time. In addition, 1,3-enynes could also be compatible with the reaction system to carry out the three-component 1,4-alkylcyanation reaction. The reaction exhibits high chemo- and regioselectivity and wide substrate scope and provides an efficient route to the synthesis of carbonyl-containing alkyl nitriles and allenyl nitriles in a single step. Mechanistic studies also prove that sulfoxonium ylides serve as carbon-centered radical precursors and will inspire new opportunities for the design of radical-mediated multicomponent reactions. Further application of sulfur ylides is under investigation in our lab.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c03998.

Experimental procedures, full spectroscopic data, crystal data, mechanism studies, figures of emission quenching, and ^{1}H and ^{13}C NMR spectra of all compounds (PDF)

Accession Codes

CCDC 2369428 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: + 44 1223 336033.

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Notes

The authors declare no competing financial interest.

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